

Power BI

"Empowering Decisions: The Power of Power BI"





what exactly is Business Intelligence, anyway?

business intelligence is all about leveraging data to make better decisions. This can take many forms and is not necessarily restricted to just business. We use data in our personal lives to make better decisions as well,

For example, if we are remodeling a bathroom, we get multiple quotes from different firms. The prices and details in these quotes are pieces of data that allow us to make an informed decision in terms of which company to choose



EXPLORING KEY CONCEPTS OF BUSINESS INTELLIGENCE

In the context of organizations, business intelligence is about making better decisions for your business.

what can make their business more effective, efficient, and profitable

- **How can the business attract new customers?**
- **How can the business retain more customers?**
- **Who are the competitors and how do they compare?**
- **What is driving profitability?**
- **Where can expenses be diminished?**

key concepts of business intelligence
can be broken into five areas



DOMAIN

DATA

MODEL

ANALYSIS

VISUALIZATION

DOMAIN

A **domain** is simply the context where business intelligence is applied. Most businesses are composed of relatively standard business functions or departments

- Sales
- Marketing
- Manufacturing/production
- Supply chain/operations
- Research and development
- Human resources
- Accounting/finance



DATA



Identifying the sources of relevant data. These sources may be internal or external to an organization and may be structured, unstructured, or semi-structured in nature

- **Internal Data** (within an Organisation)
- **External Data** (generates outside organisation)
- **Structured Data** (specific table with rows and columns)
- **UnStructured Data** (cannot organised with table)
- **Semi-Structured Data** (like html,xml,json..)

MODEL

A model, or data model, refers to the way in which one or more data sources are organized to support analysis and visualization. Models are built by transforming and cleansing data, helping to define the types of data within those sources

- **Organizing**
- **Transforming and cleansing**
- **Defining and categorizing**

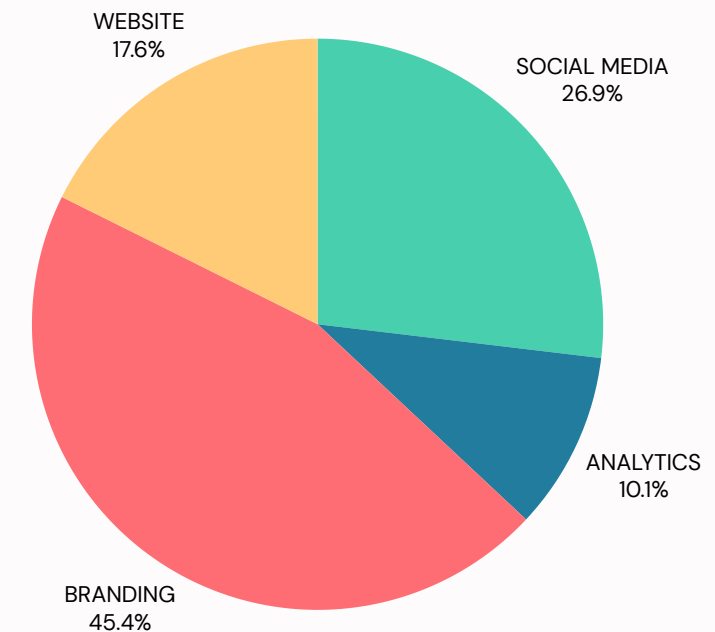
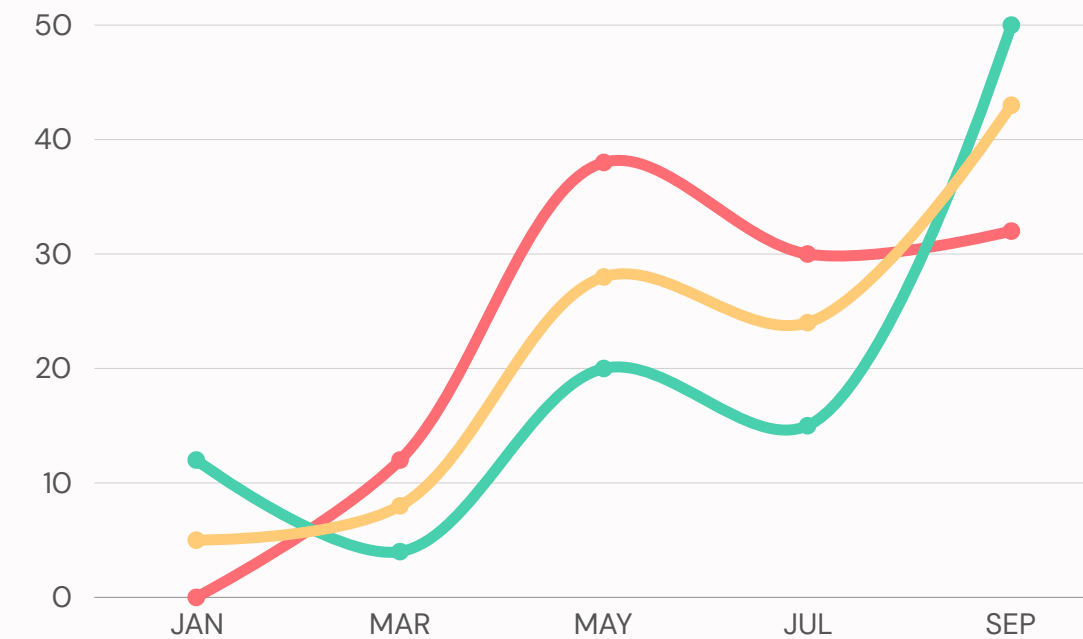
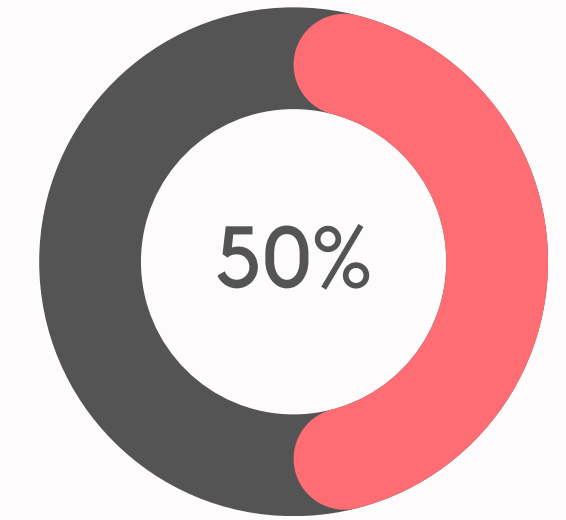
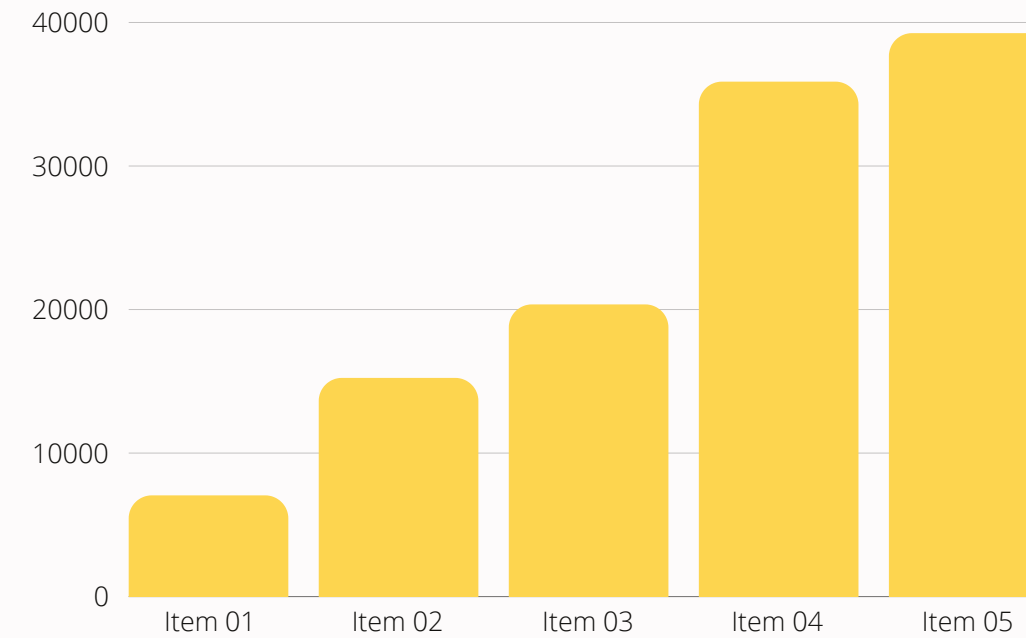
Analysis

For example, to **predict future sales** revenue, it is important that such data is **aggregated and analyzed**. This analysis can determine the **average sales** for a product, the frequency of purchases, and which customers purchase more frequently than others. Such information allows **better decision-making** by an organization

VISUALIZATION

The final key concept in business intelligence is visualization or the actual presentation of the analysis being performed

results of the analysis in the form of charts, reports, and dashboards. This may take the form of tables, matrices, pie charts, bar graphs, and other visual displays that help provide context and meaning to the analysis



MEET POWER BI



Microsoft Power BI is a self-service business intelligence platform, which includes both desktop and web-based applications for connecting, modeling, and visualizing data

Learn more at powerbi.microsoft.com

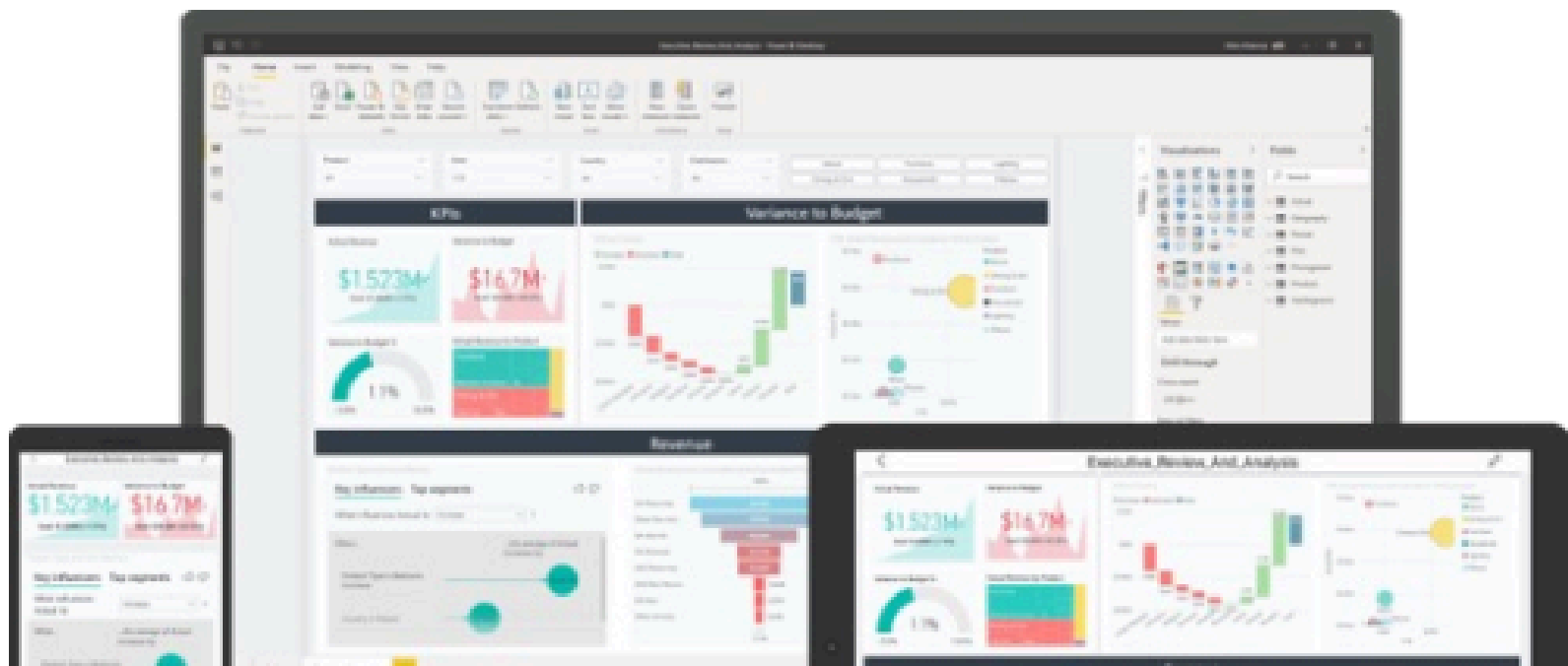


Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms



WHY POWER BI?



Connect, transform and load millions of rows of data

- *Access data from virtually anywhere (database tables, flat files, web, cloud services, folders, etc.), and create fully automated workflows to extract, transform and load data for analysis*



Build relational models to blend data from multiple sources

- *Create table relationships to analyze holistic performance across an entire relational data model*



Define complex calculations using Data Analysis Expressions (DAX)

- *Enhance datasets and enable advanced analytics with powerful and portable DAX expressions*



Bring data to life with interactive reports and dashboards

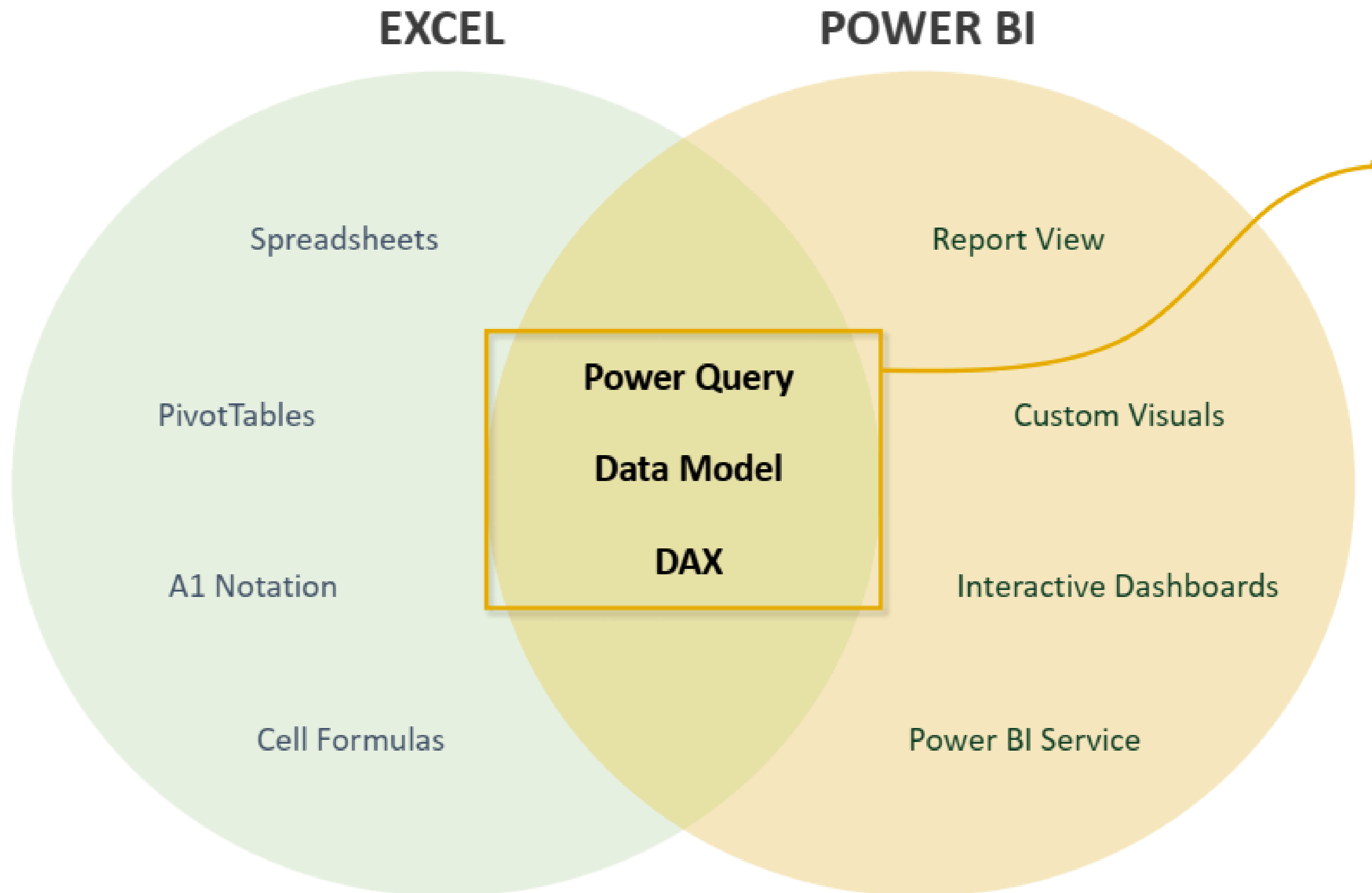
- *Build professional-quality reports and dashboards with best-in-class visualization tools*



Develop a versatile, in-demand skill set

- *Power BI is the industry leader in self-service BI, and the skills you build in this course will be highly transferrable*

EXCEL VS. POWER BI



Excel and Power BI are built on top of the **same analytics engines**

- Power BI takes the same data transformation and modeling capabilities and adds **powerful visualization and publishing tools**
- Transitioning is easy; you can import an **entire data model** directly from Excel!

POWER BI



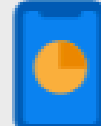
POWER BI DESKTOP

a Windows desktop
application for data
analysis and reports
creation



POWER BI SERVICE

online Software as a
Service, used to
collaborate and
distribute Power BI
reports



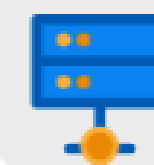
POWER BI MOBILE

a native mobile
app



POWER BI REPORT BUILDER

a tool for creating
paginated reports



POWER BI REPORT SERVER

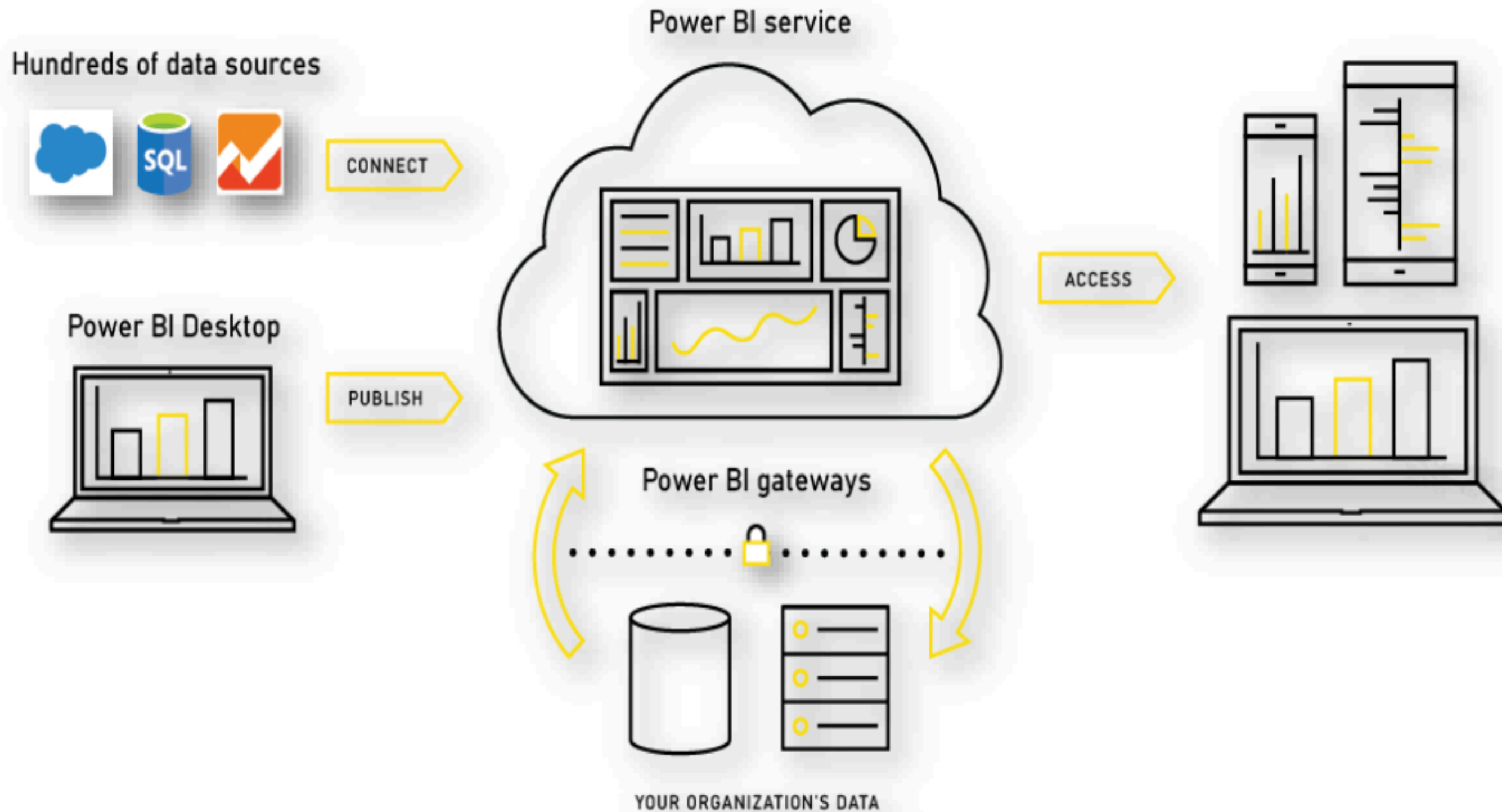
an on-premises
report server
(available through
Power BI Premium
licensing)



POWER BI EMBEDDED

a service to embed
visuals, reports, and
dashboards into an
application

Power BI Desktop Architecture



Power BI Desktop Components

Power BI Desktop uses three components to perform different operations. They are:

Power Query

Power Pivot

Power View



System Requirements and Installation

Minimum Hardware Requirements

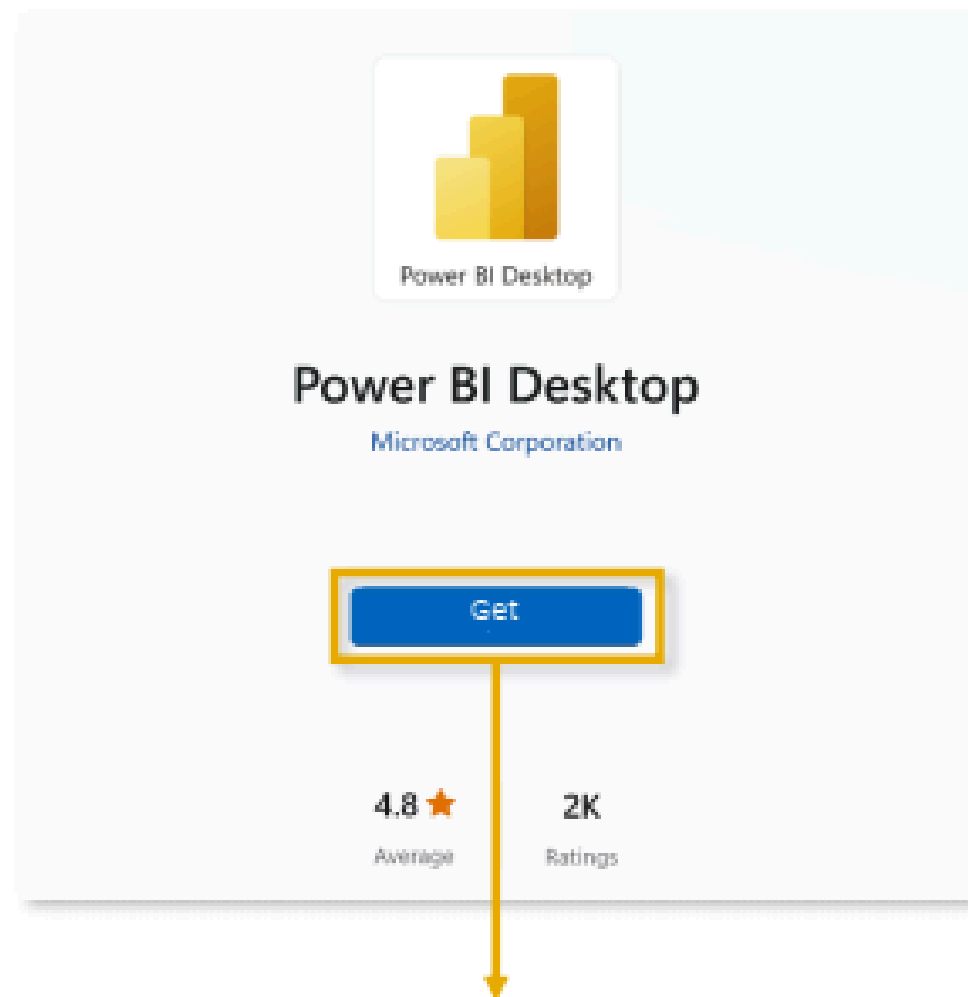
Before you install Power BI Desktop, your system should meet the minimum hardware requirements. They are:

- Windows 7 / Windows Server 2008 R2, or later
- .NET 4.5
- Internet Explorer 9 or later
- 1 gigahertz (GHz) or faster x86- or x64-bit processor
- At least 1440x900 or 1600x900 (16:9)
- Minimum 1 GB space should be available, Recommended available space should be 1.5 GB or more

INSTALLING POWER BI DESKTOP

1) Download from **Microsoft store**

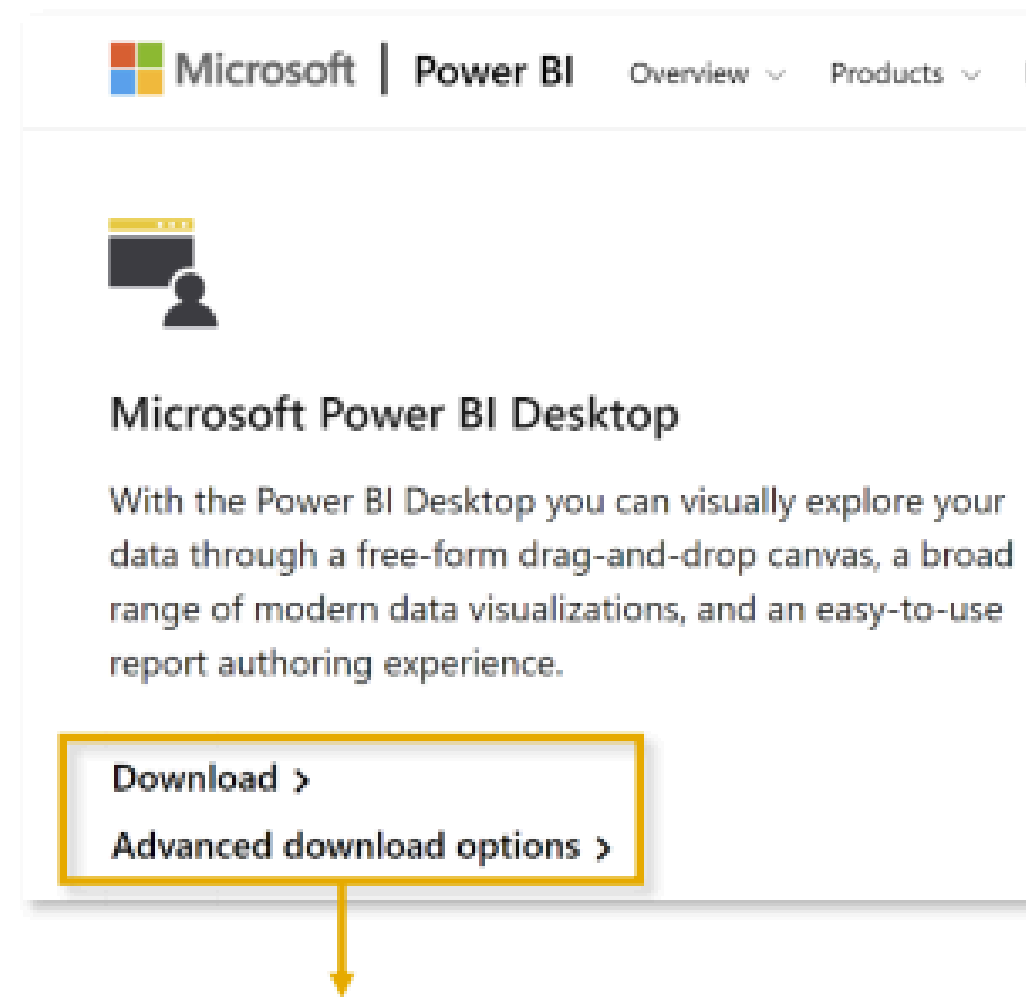
apps.microsoft.com



- Windows handles **automatic updates**
- Updates only elements that have been changed
- Doesn't require administrator access

2) Download **manually from web**

powerbi.microsoft.com/downloads



- **No automatic updates** (allows version control)
- Downloads an executable installation file
- Administrator access may be required

3) Install as part of **Microsoft 365**

microsoft.com/en-us/microsoft-365



- Power BI Desktop is included as part of select enterprise Office/Microsoft 365 subscriptions
- If your company uses a compatible version of Microsoft 365, talk to an admin about getting access to Power BI

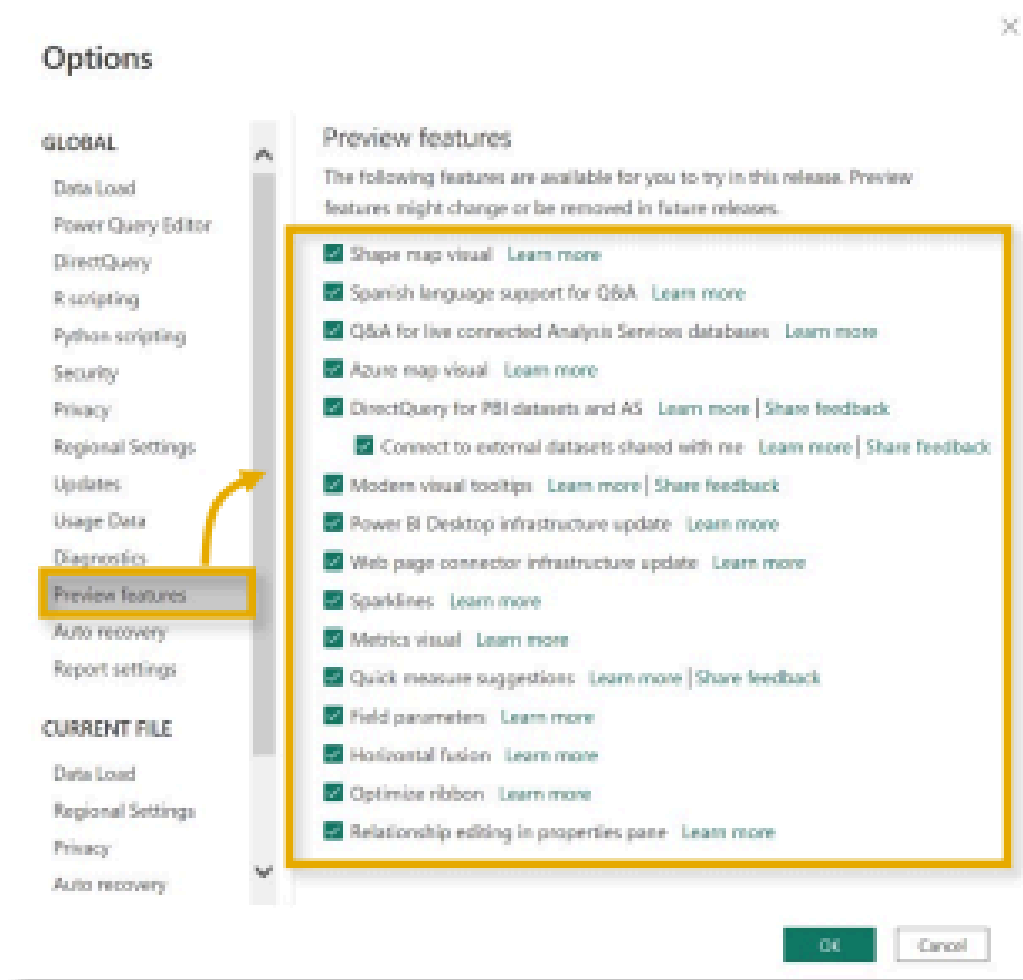


HEY THIS IS IMPORTANT!

You do **NOT** need to register for a Power BI Pro account to access Power BI Desktop

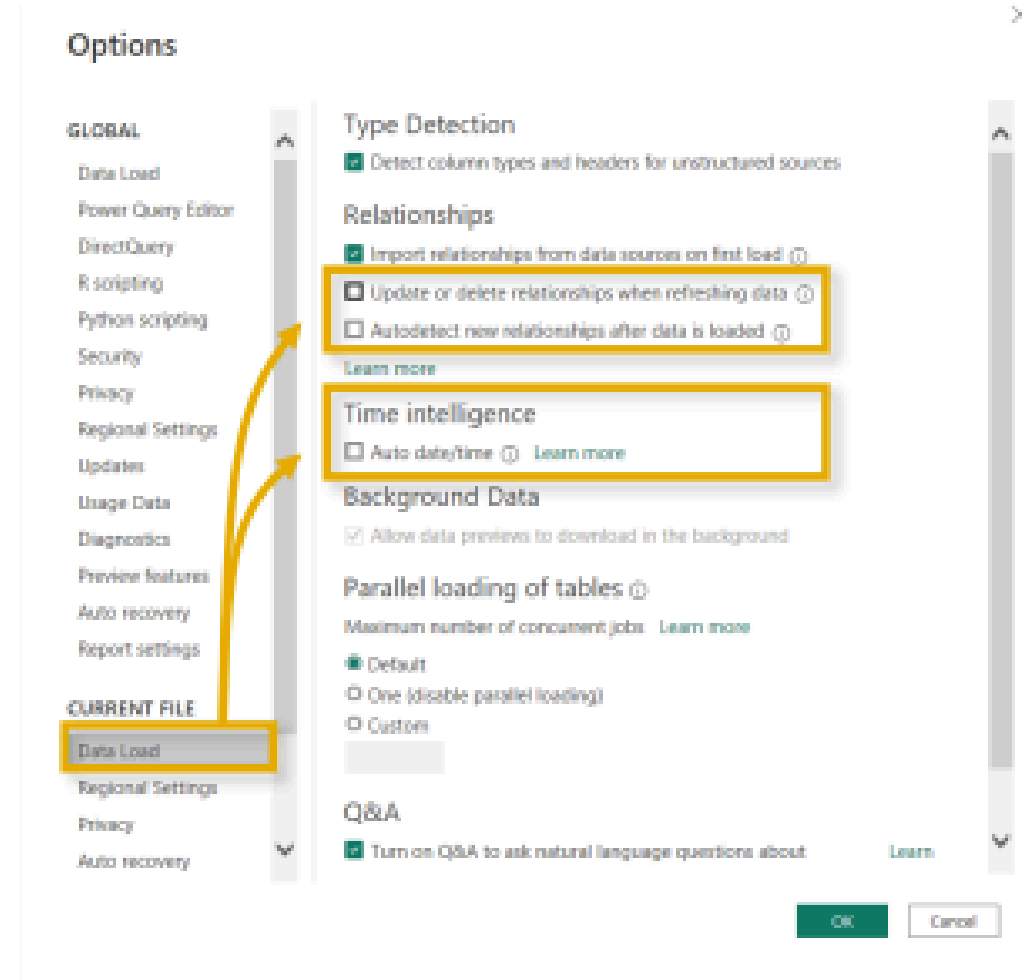
POWER BI SETTINGS

Global > Preview Features



Select **all available preview features** by default (these change with each monthly release)

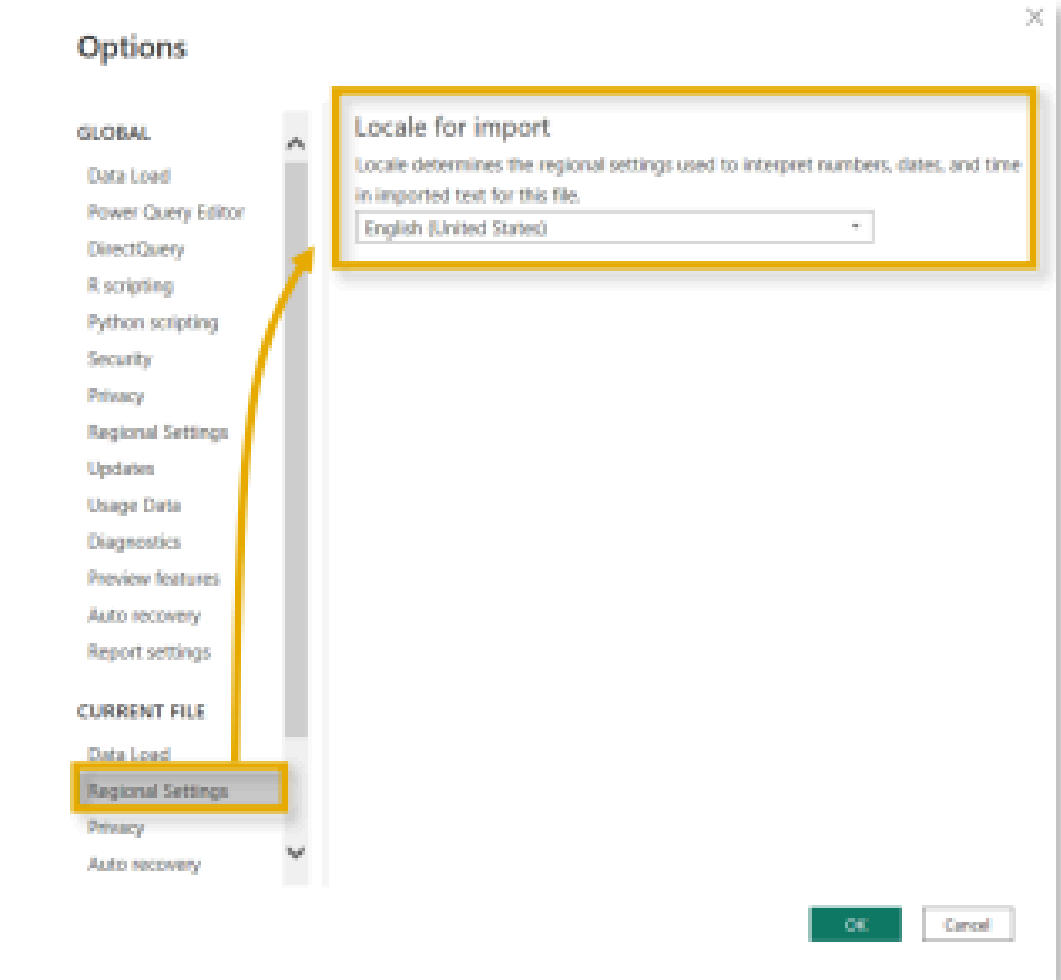
Current File > Data Load



Make sure the following options are **NOT** selected:

- **Update or delete relationships when refreshing data**
- **Autodetect new relationships after data is loaded**
- **Time Intelligence > Auto date/time**

Current File > Regional Settings

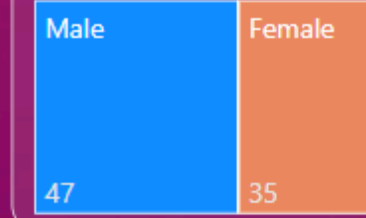


Select **"English (United States)"** from the dropdown menu (this will align with the data in course project files)

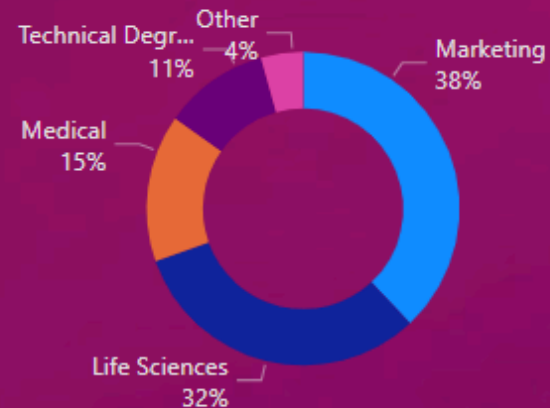


HEY THIS IS IMPORTANT!

Options under **CURRENT FILE** need to be adjusted **every time you open a new Power BI workbook** (these settings do not persist across new .pbix files)



7.3



Age Group	Percentage
26-35	41
18-25	18
46-55	17
36-45	16
55+	0

JobRole	1	2	3	4	Total
Manager	0	1	0	1	2
Sales Executive	16	9	18	14	57
Sales Representative	7	10	9	7	33
Total	23	20	27	22	92

Budget Range	Number of Projects
Upto 5k	43
5k-10k	36
10k-15k	11
15k+	2

Age Group	Number of People
0-1	7
2-3	17
4-5	12
6-7	10
8-9	7
10-11	7
12-13	5
14-15	2
16-17	5
18-19	4
20-21	4
22-23	2

Job Title	Count
Sales Executive	57
Sales Representative	33
Manager	2

Filters

 Search

✓  HR_Analytics

- ☐ Σ Age
- ☐ AgeGroup
- ☐ Attrition
- ☐ Σ AttritionCount
- ☐  AttritionRate
- ☐ BusinessTravel
- ☐ Σ DailyRate
- ☐ Department
- ☐ Σ DistanceFromH...
- ☐ Σ Education
- ☐ EducationField
- ☐ EmpID
- ☐ Σ EmployeeCount
- ☐ Σ EmployeeNum...
- ☐ Σ EnvironmentSa...
- ☐ Gender
- ☐ Σ HourlyRate
- ☐ Σ JobInvolvement
- ☐ Σ JobLevel
- ☐ JobRole
- ☐ Σ JobSatisfaction
- ☐ MaritalStatus

Add data fields here

Drill through

Cross-report

Keep all filters

Add drill-through fields here